

lab02

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1.

a) números de voos com atraso > 10 , o número total de voos em cada grupo com essas duas coisas podemos combina-las de varios lotes somando-as e então calculando a porcentagem final

2.

```
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

`filter`, `lag`

Os seguintes objetos são mascarados por 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(readr)

getStats <- function(input, pos) {
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%
    filter(!is.na(ARRIVAL_DELAY)) %>%
    group_by(YEAR, MONTH, DAY, AIRLINE) %>%
    summarise(
      atrasos = sum(ARRIVAL_DELAY > 10),
      total = n(),
```

```

      .groups = "drop"
    )
  }

```

3.

```

resultado_bruto <- read_csv_chunked(
  "flights.csv",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = cols_only(
    YEAR = col_integer(),
    MONTH = col_integer(),
    DAY = col_integer(),
    AIRLINE = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

```

4.

```

computeStats <- function(stats) {
  stats %>%
    group_by(AIRLINE, YEAR, MONTH, DAY) %>%
    summarise(
      atrasos = sum(atrasos),
      total = sum(total),
      .groups = "drop"
    ) %>%
    mutate(
      Data = as.Date(paste(YEAR, MONTH, DAY, sep = "-")),
      Perc = atrasos / total
    ) %>%
    rename(Cia = AIRLINE) %>%
    select(Cia, Data, Perc)
}

resultado <- computeStats(resultado_bruto)
resultado

```

A tibble: 1,276 x 3

	Cia	Data	Perc
	<chr>	<date>	<dbl>
1	AA	2015-01-01	0.373
2	AA	2015-01-02	0.430
3	AA	2015-01-03	0.538
4	AA	2015-01-04	0.602
5	AA	2015-01-05	0.491
6	AA	2015-01-06	0.366
7	AA	2015-01-07	0.355
8	AA	2015-01-08	0.307
9	AA	2015-01-09	0.342
10	AA	2015-01-10	0.280

i 1,266 more rows

5.

```
library(ggcal)
library(ggplot2)

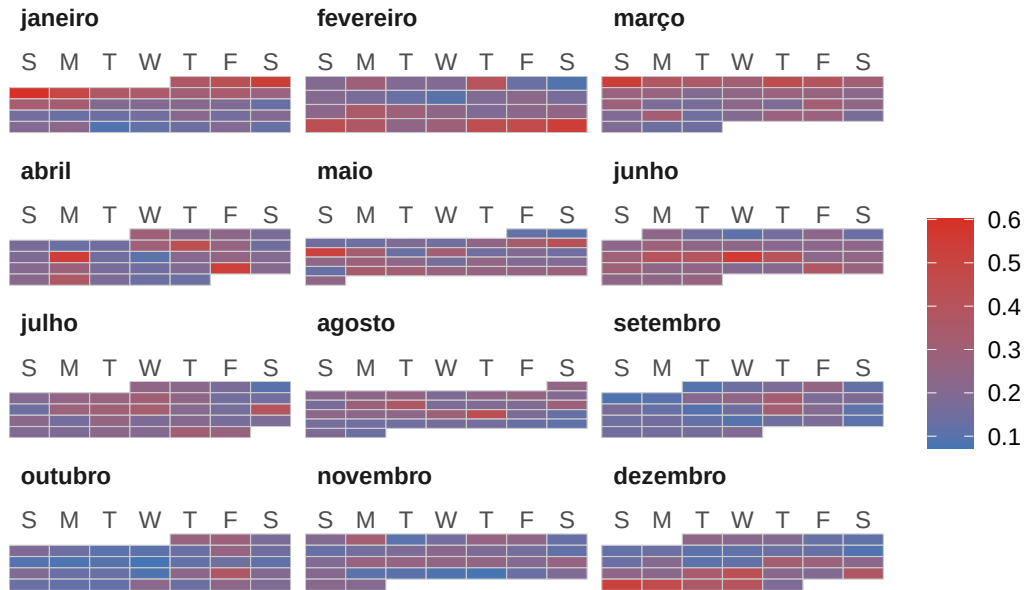
pal <- scale_fill_gradient(low = "#4575b4", high = "#d73027")

baseCalendario <- function(stats, cia) {
  sub <- stats %>% filter(Cia == cia)
  ggcal(sub$Data, sub$Perc)
}

cAA <- baseCalendario(resultado, "AA")
cDL <- baseCalendario(resultado, "DL")
cUA <- baseCalendario(resultado, "UA")
cUS <- baseCalendario(resultado, "US")

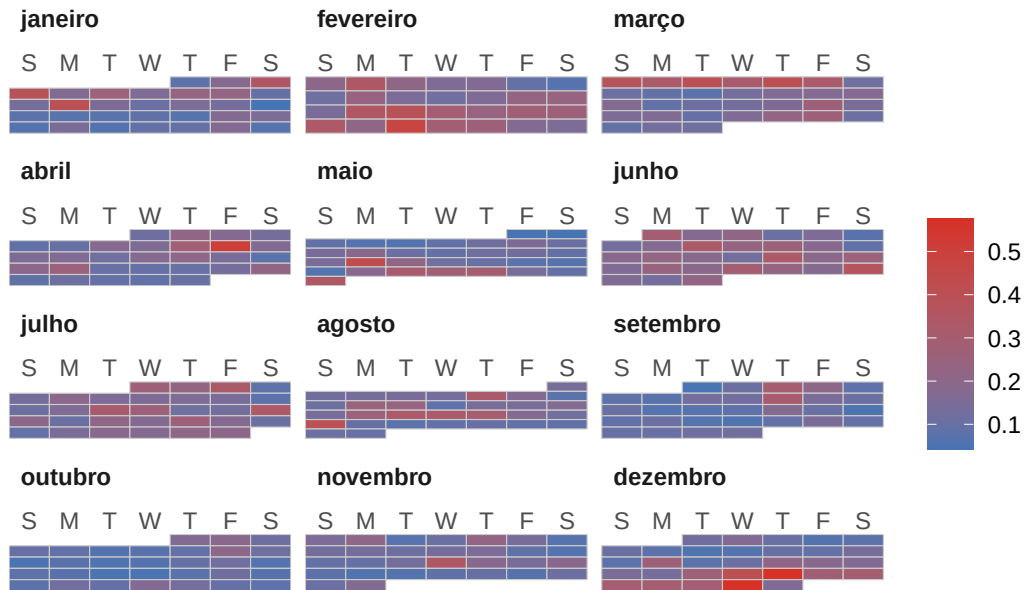
cAA + pal + ggtitle("American Airlines")
```

American Airlines



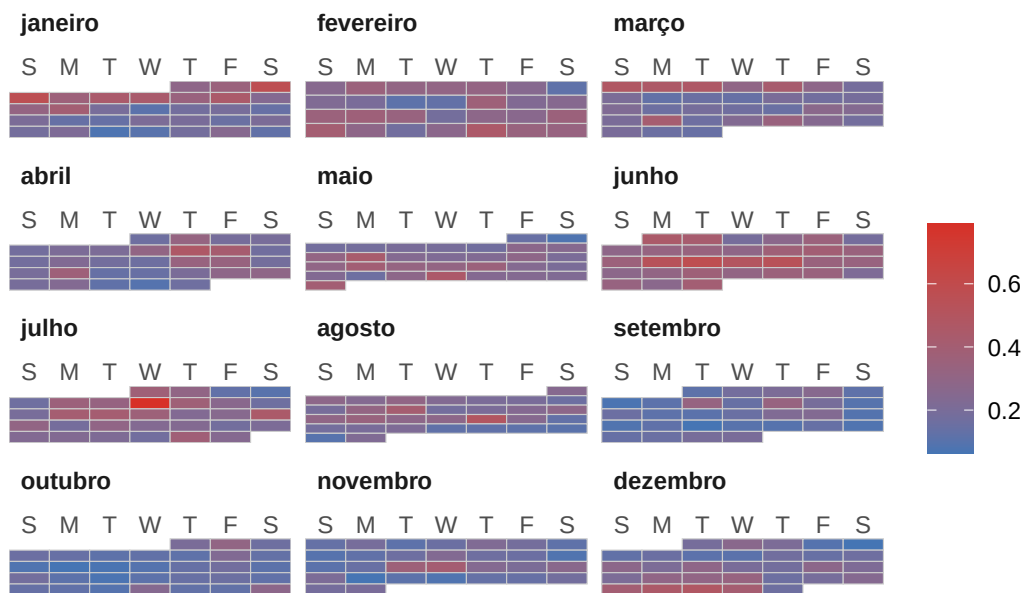
```
cDL + pal + ggtitle("Delta Airlines")
```

Delta Airlines



```
cUA + pal + ggtitle("United Airlines")
```

United Airlines



```
cUS + pal + ggtitle("US Airways")
```

US Airways

